Quiz II

Test your knowledge of different systems that you have learned in this course by completing this quiz.

Quiz ?

Statement

In Google File System (GFS), when a client needs to access data, it contacts the master and gets the data from the master.

Select Multiple Options

False ⊗
☐ True

Correct ✓
☑ False

Check Answer

Correct Answer ✓

Quiz ?

Statement

The main responsibility of the GFS master is:

Select Multiple Options

False ⊗
☐ To coordinate all read and write requests from clients.

False ⊗
☐ To balance the storage load between different ChunkServers.

False ⊗
☐ To coordinate concurrent write requests to a file from multiple clients.

Correct ✓
☑ Store all the filesystem metadata, like file names, total parts, locations of the parts, etc.

Check Answer

Correct Answer ✓

Quiz ?

Statement

When scaling systems, mostly distributed system give up:

Select Multiple Options

False ⊗
☐ Availability in favor of Partition-tolerance

False ⊗
☐ Availability in favor of Consistency

False ⊗
☐ Atomicity in favor of Availability

Correct ✓
☑ Data consistency in favor of availability

Check Answer

Correct Answer ✓

Quiz ?

Statement

The motivation for an eventual consistency model was:

Select Multiple Options

Correct ✓
☑ It is impossible to have highly available replicated data that is fully consistent in a system that can survive network partitioning.

False ⊗
☐ Data inconsistencies are highly undesirable, so the primary focus should be on ensuring that all replicas are consistent.

False ⊗
☐ Distributed commit protocols can never work reliably, so data is bound to become inconsistent on some participants.

False ⊗
☐ Data might be in an inconsistent state during the execution of transactions but will be made consistent when they commit.

Check Answer

Correct Answer ✓

Quiz ?

Statement

In BigTable, a Tablet refers to:

Select Multiple Options

False ⊗
☐ A subset of rows and column families

False ⊗
☐ A contiguous range of rows of a table

False ⊗
☐ All columns of contiguous rows of a Locality Group

Correct ✓
☑ All columns families of contiguous rows of a table

Check Answer

Correct Answer ✓

Quiz ?

Statement

Split-brain is a condition that arises when:

Select Multiple Options

False ⊗
☐ A leader election is taking place.

Correct ✓
☑ A network is partitioned.

False ⊗
☐ The election algorithm fails to elect a leader.

False ⊗
☐ The leader initiates a leader election.

Check Answer

Correct Answer ✓

Quiz ?

Statement

The main difference between the Google File System (GFS) and the Hadoop Distributed File System (HDFS) is that HDFS:

Select Multiple Options

False ⊗
☐ HDFS stores all filesystem metadata on a single master.

False ⊗
☐ HDFS stores files in fixed-size blocks.

False ⊗
☐ HDFS does not allow concurrent reading of a file.

Correct ✓
☑ In HDFS, there are no concurrent writers and arbitrary file modifications.

Explanation

Contrary to GFS, multiple writers cannot concurrently write to an HDFS file. Furthermore, writes are always made at the end of the file, in an append-only fashion. There is no support for modifications at arbitrary offsets in a file.

Check Answer

Correct Answer ✓

Quiz ?

Statement

Distributed systems use the following techniques for Fencing. Select all that apply.

Select Multiple Options

False ⊗
☐ Resource Fencing

False ⊗
☐ Network Fencing

Correct ✓
☑ Node Fencing

False ⊗
☐ None of the above

Check Answer

Correct Answer ✓

Quiz ?

Statement

In distributed systems, Fencing is used to:

Select Multiple Options

False ⊗
☐ To isolate a central server from other servers.

False ⊗
☐ To isolate a server while it initializes.

Correct ✓
☑ Isolate a faulty server to prevent it from doing any damage or causing corruption.

Explanation

By means of Fencing, we take a faulty server out of service.

False ⊗
☐ To enhance security around certain critical nodes.

Check Answer

Correct Answer ✓

Quiz ?

Statement

A column family in BigTable is:

Select Multiple Options

False ⊗
☐ A group of columns within a single table.

Explanation

A column family is a storage unit for multiple columns in a table – usually related in function. From a performance point, data in a row is retrieved by column families.

False ⊗
☐ A group of related columns among multiple tables.

Correct ✓
☑ A set of columns that are shared across multiple tables.

False ⊗
☐ A preconfigured set of columns to make it easier to set up a Bigtable instance.

Check Answer

Correct Answer ✓

Quiz ?

Statement

Dynamo uses vector clocks to:

Select Multiple Options

False ⊗
☐ To garbage collect older versions of an object.

False ⊗
☐ To store multiple versions of data.

Correct ✓
☑ Determine causality between different updates on the same object.

Explanation

Dynamo uses vector clocks to find causality for concurrent updates on an object.

False ⊗
☐ To serialize concurrent updates to an object.

Check Answer

Correct Answer ✓

Input your markdown text here

Quiz ?

Statement

In distributed systems, replication gives following benefits. Select all that apply.

Select Multiple Options

Correct ✓
☑ Availability

False ⊗
☐ Consistency

Correct ✓
☑ Reliability

Explanation

Dynamo uses vector clocks to find causality for concurrent updates on an object.

Correct ✓
☑ Better read performance

Check Answer

Correct Answer ✓

Quiz ?

Statement

In distributed systems, Heartbeat is used to:

Select Multiple Options

Correct ✓
☑ Detect dead servers

False ⊗
☐ Synchronize operations between servers

False ⊗
☐ Measure load on servers

False ⊗
☐ Transfer configuration changes

Check Answer

Correct Answer ✓

Quiz ?

Statement

What statements are true about the Google File System (GFS)? Select all that apply.

Select Multiple Options

Correct ✓
☑ Different parts of a file can be stored on different servers.

Correct ✓
☑ GFS can store multiple replicas of a file's data on different servers.

Correct ✓
☑ Any file stored in GFS can be bigger than the capacity of a server.

False ⊗
☐ Transfer configuration changes

Check Answer

Correct Answer ✓

Quiz ?

Statement

**What is the importance of choosing a good row key in BigTable? **

Select Multiple Options

Correct ✓
☑ Take advantage of the locality of adjacent rows.

Explanation

BigTable keeps its data sorted by the row key. Rows are stored next to each other in a tablet – except occasionally when we have to jump to the next tablet.

False ⊗
☐ Avoid the use of duplicate keys.

False ⊗
☐ Enable rapid search for an arbitrary row based on its key.

False ⊗
☐ Make queries user-friendly for easier debugging.

Check Answer

Correct Answer ✓

← Previous
Quiz I

Next →
Contact Us

☐ Mark as Completed